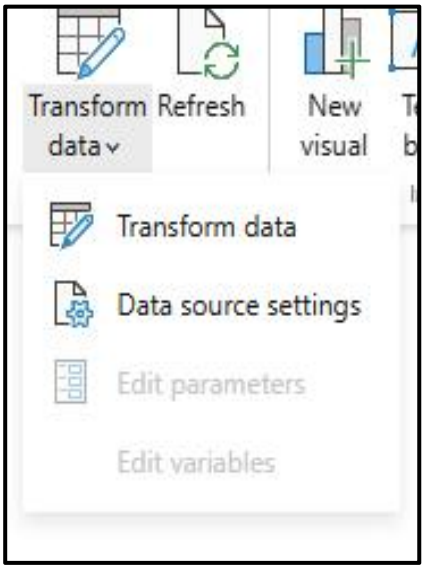
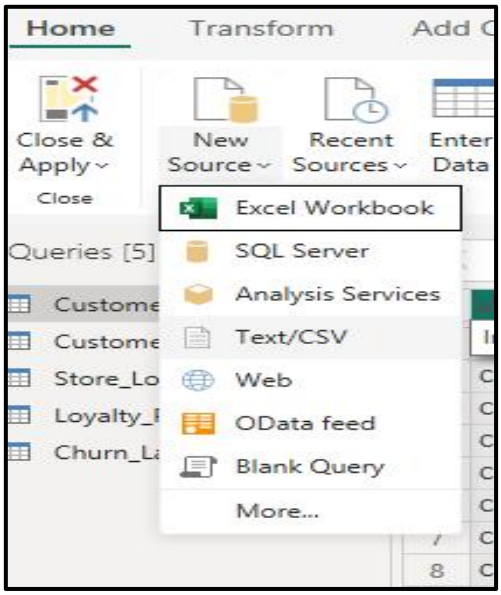
POWER BI PROJECT

Task 1 – Data Modeling and Cleaning

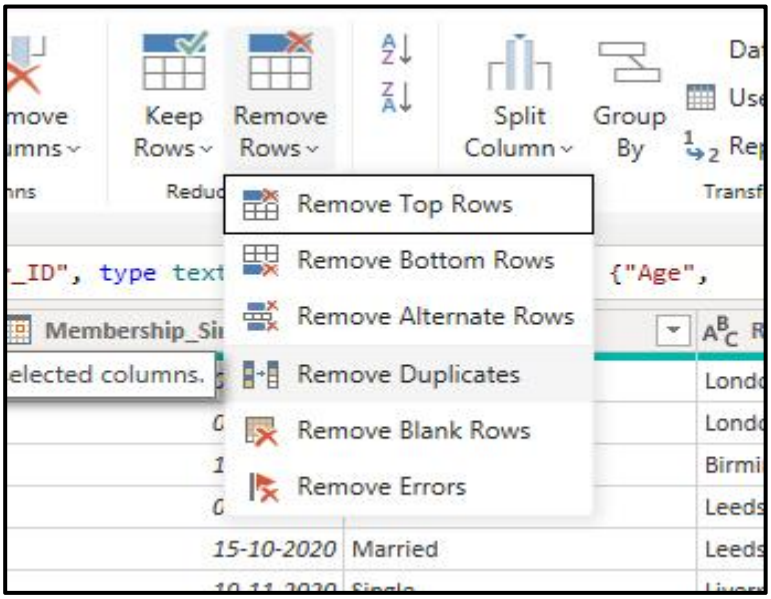
- **To remove duplicate rows based on key columns (e.g. – Customer Id, Store Id etc.)**

Step 1 – To load data in power query editor, click on **transform data** in **Home Tab** of Power BI user interface then power query editor window will open, then from **Home Tab** click on **New Resource** choose file type (e.g. – Excel Workbook, Text/CSV etc.) and load it.





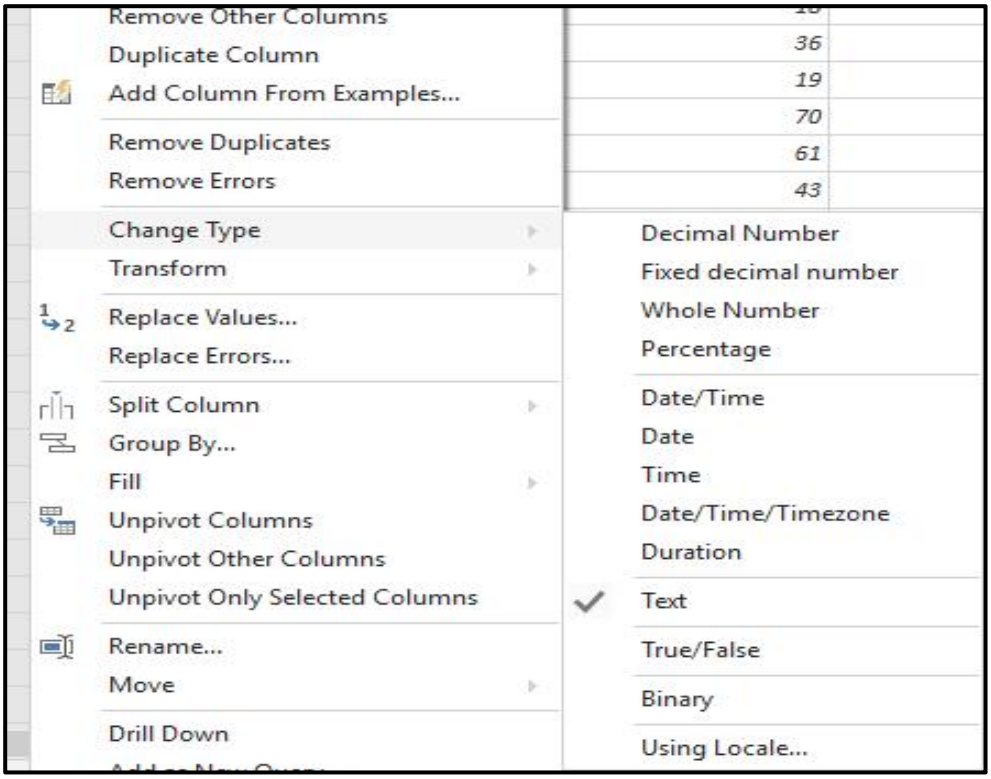
Step 2 – To remove duplicate rows, select the key column from data table and in **Home Tab** click on **Remove Rows** and from dropdown list click on **Remove Duplicates** it will remove all duplicate rows from table based on the selected column. Do same for other key columns.



- **To Format Columns: convert dates, ensure numeric types for amount, points, etc.**

Step 1 – To format column in power query editor, select the column and right click.

Step 2 - Now, choose **Change Type** and select the suitable data type for selected column, also you can change data type by clicking on icon beside column name in data table.



- **<u>To handle missing value and null appropriately.</u>**

To handle missing or null value we have to go to **Power Query Editor**
In Pwer Query Editor home tab we can use **Replace Value** to replace value for null with some appropirate value and for unique value always use **Filter**.

For **Customer IDs, Transaction IDs** → never replace, always filter (since IDs must be unique & valid).

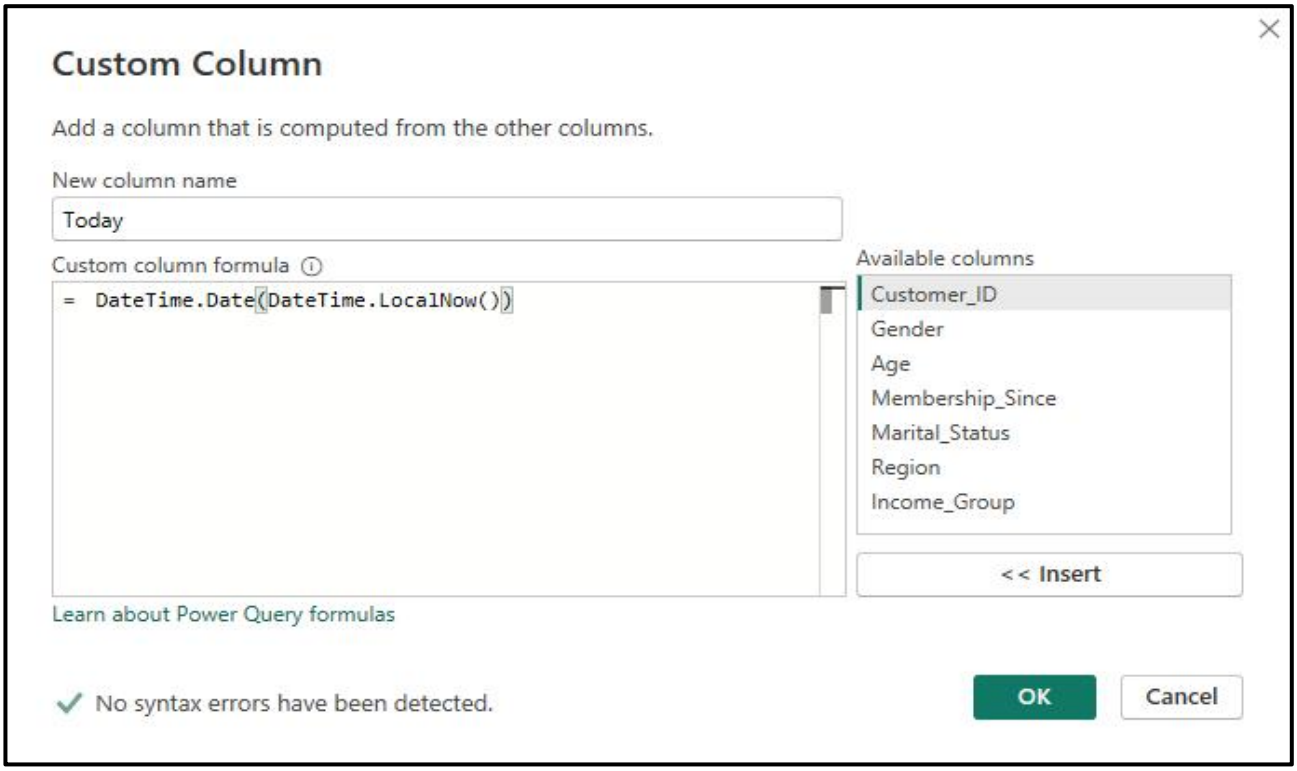
For **Amount** → replace with 0.

For **Categorical columns (Region, Store Type)** → use "Unknown" to avoid dropping customers.

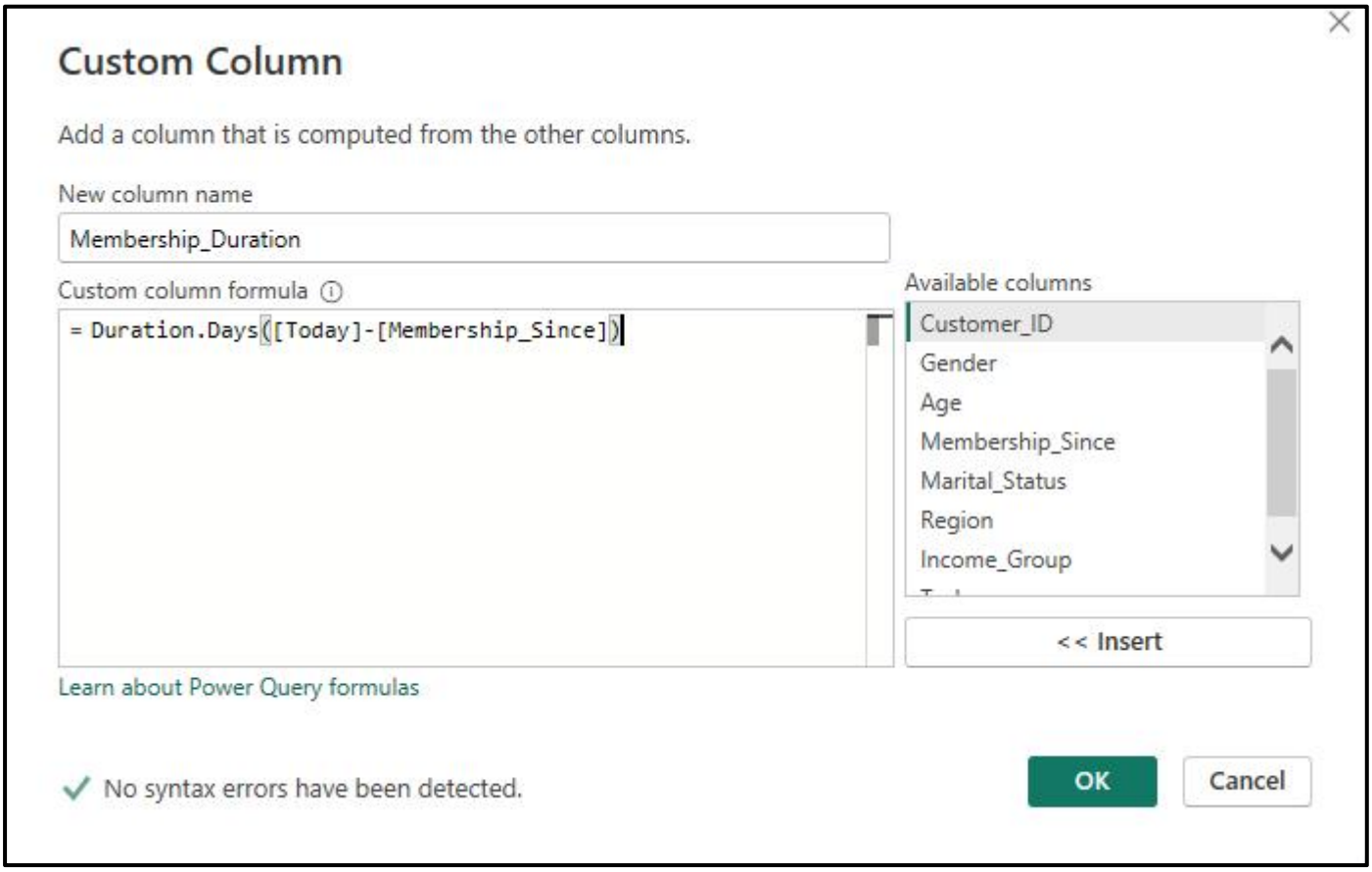
- **<u>To create column Membership Duration = Today – Membership Since</u>.**

Step 1 - To create a new column having today's date in power query editor go to **<u>Add Column Tab</u>** and select **<u>Custom Column.</u>**

Step 2 – Give name Today to the new column and use custom column formula =**DateTime.Date(DateTime.LocalNow())** and press Ok, a new column named as Today containing today's date will be added to data table.



Step 3 – Now again go to Add Column Tab and click on Custom Column and name the column as Membership Duration (Days) and use custom column formula = Duration.Days([Today] – [Membership_Since]) and click Ok, a new column named as Membership duration (Days).



Today	Membership_Duration
21-08-2025	1754
21-08-2025	1508
21-08-2025	1464
21-08-2025	567
21-08-2025	1771
21-08-2025	1736
21-08-2025	1500
21-08-2025	889
21-08-2025	1164
21-08-2025	441
21-08-2025	841
21-08-2025	1506
21-08-2025	906
21-08-2025	847
21-08-2025	579

- **To create column for Transaction Year and Transaction Month from Transaction Date column.**

Step1 – Select the Transaction Date column go to **Add Column Tab** and click on **Date** and from drop down list select **Name of Month** a new column containing transaction month will added to data table and named the column as Transaction Month.

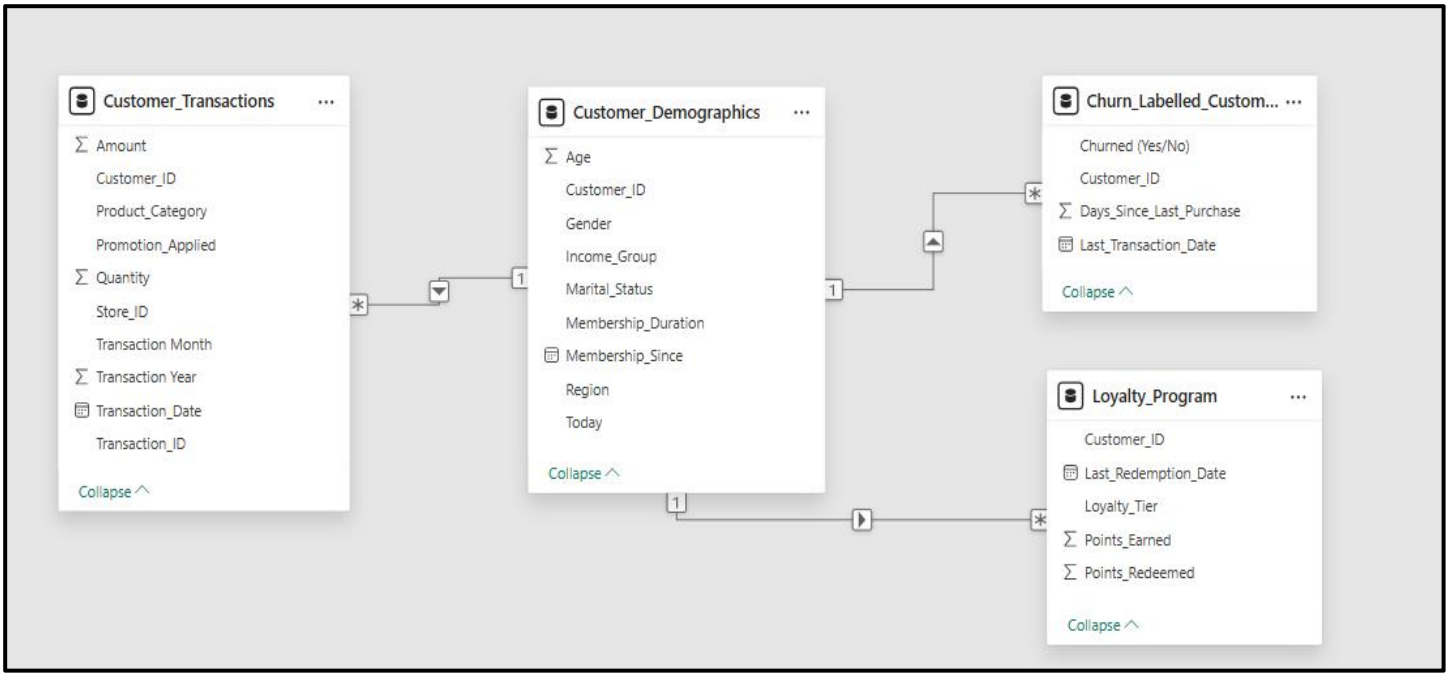
Step 2 – Similarly for Transaction Year just select the **Year** from drop down list and a new column containing transaction will be added and named as Transaction Year.

Transaction Month	Transaction Year
November	2024
May	2025
August	2024
August	2024
January	2025
July	2024
January	2025
September	2024
June	2025
June	2025
March	2025
June	2024
August	2024
May	2025
April	2025
August	2024
December	2024

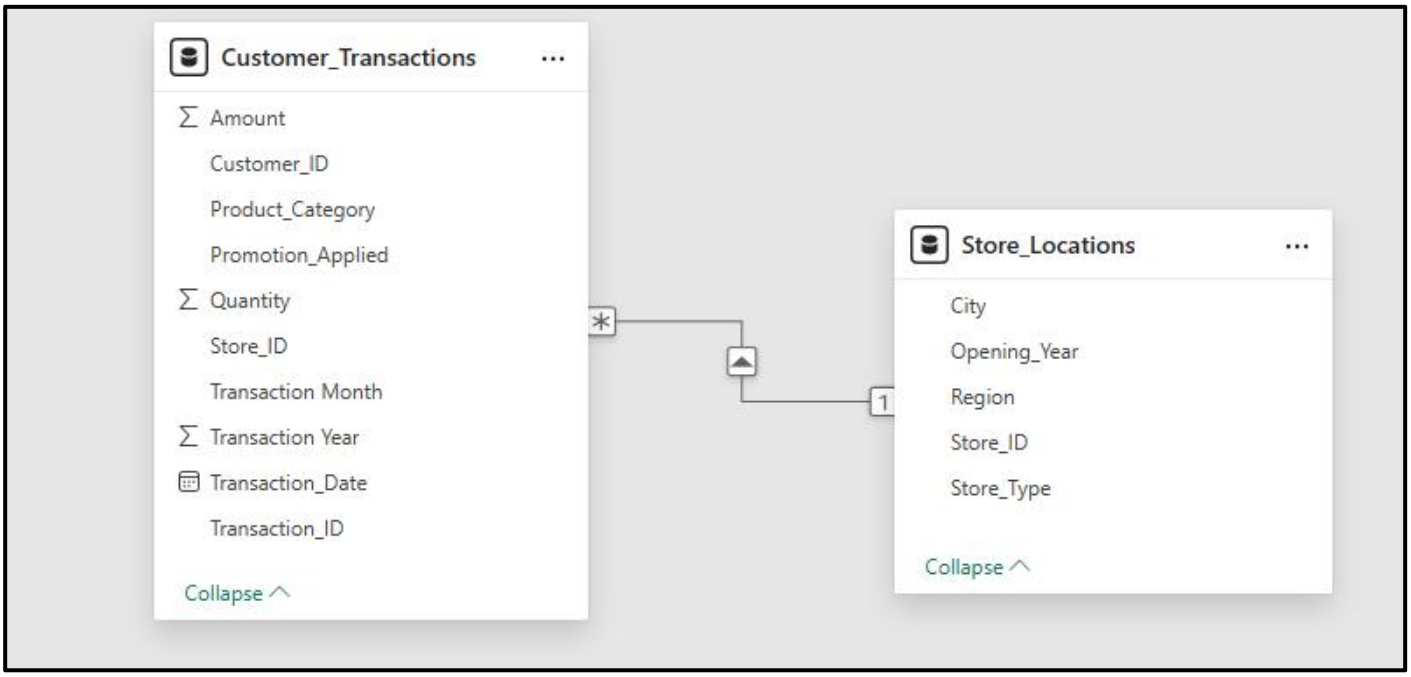
We can do that same task using **Column Using Example** in **Add Column**.

- **To create Data Model View for - One-to-Many: Customer Demographics → Transactions, Loyalty Program, Churn Labelled Customers.**

Step 1 – Load data to Power BI and go to model view and assign relation ship as follows-



- **<u>To create Data Model View for - Many-to-One: Transactions → Store_Locations</u>**



Task 2 : Churn And Retention Metrics

- **<u>Create Churn Rate Card = (Churned Customer / Total Customer)*100</u>**

To create Churn Rate Card first we have to create New Measure for total customers , churned customer and churn rate.

To create New Measure right click on **Churn Labelled Customers** table in **Data Pane** and click on New Measure.

After that write DAX for the measures as follow –

Churned Customer =CALCULATE(COUNTROWS(Churn_Labelled_Customers),'Churn_Labelled_Customer'

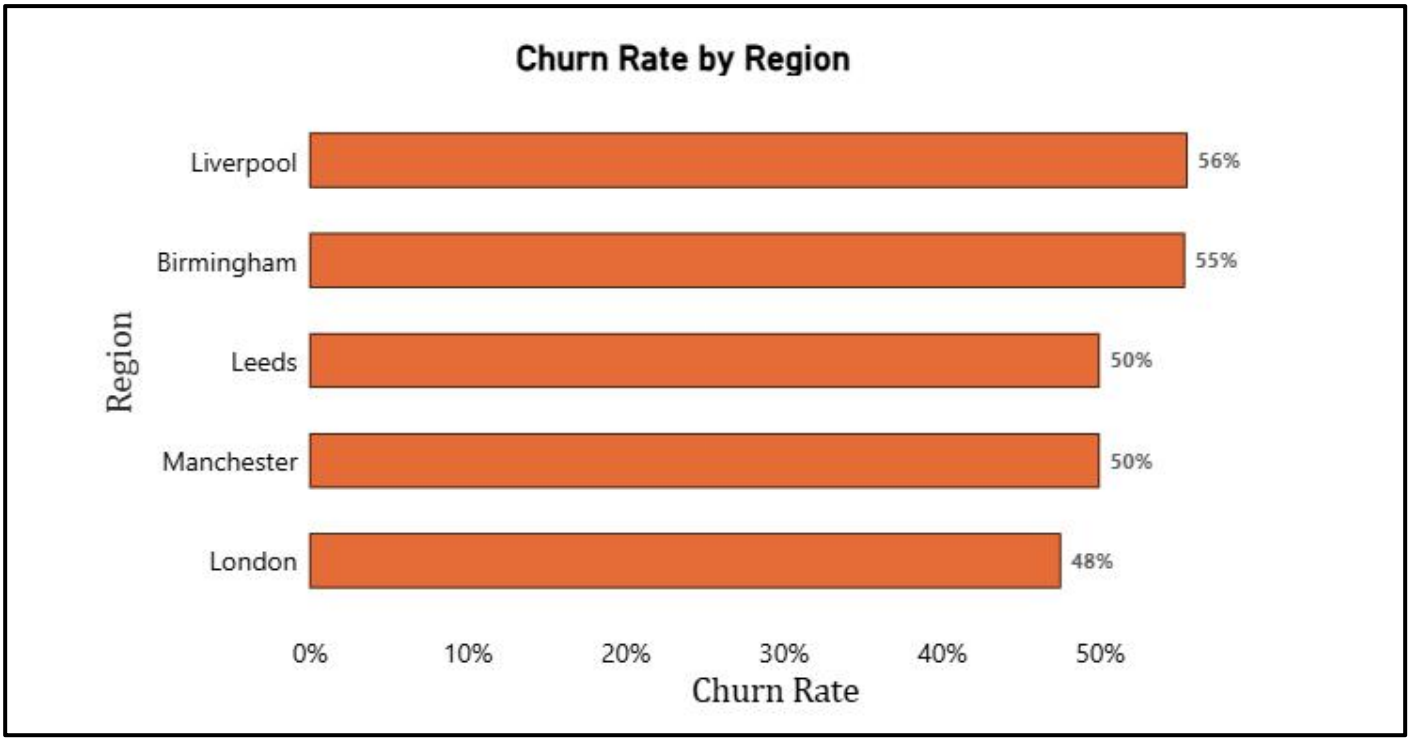
[Churned (Yes/No)] = "Yes")

Total Customers = COUNTA('Churn_Labelled_Customers'[Churned (Yes/No)])

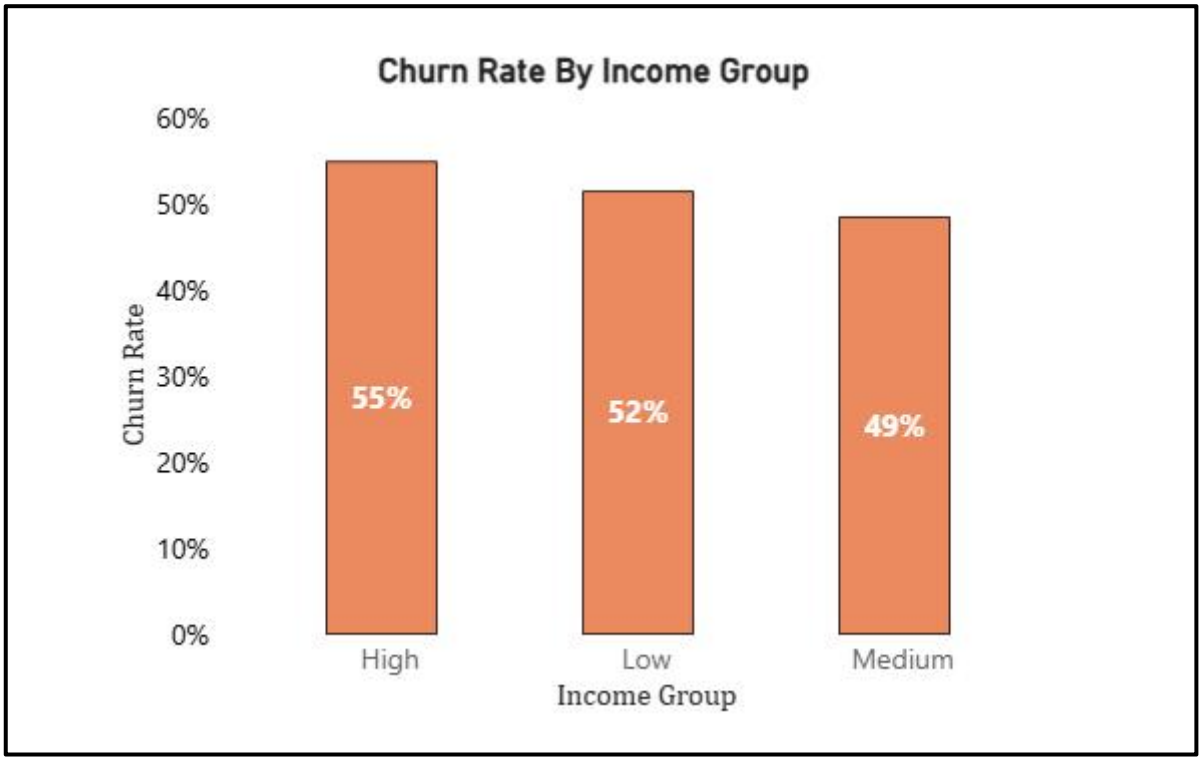
Churn Rate = [Churned Customer] / [Total Customer]

Then Measure will add to data table and select them and visualize then as **Card**.

- **<u>To visualize Churn Rate by Region –</u>**

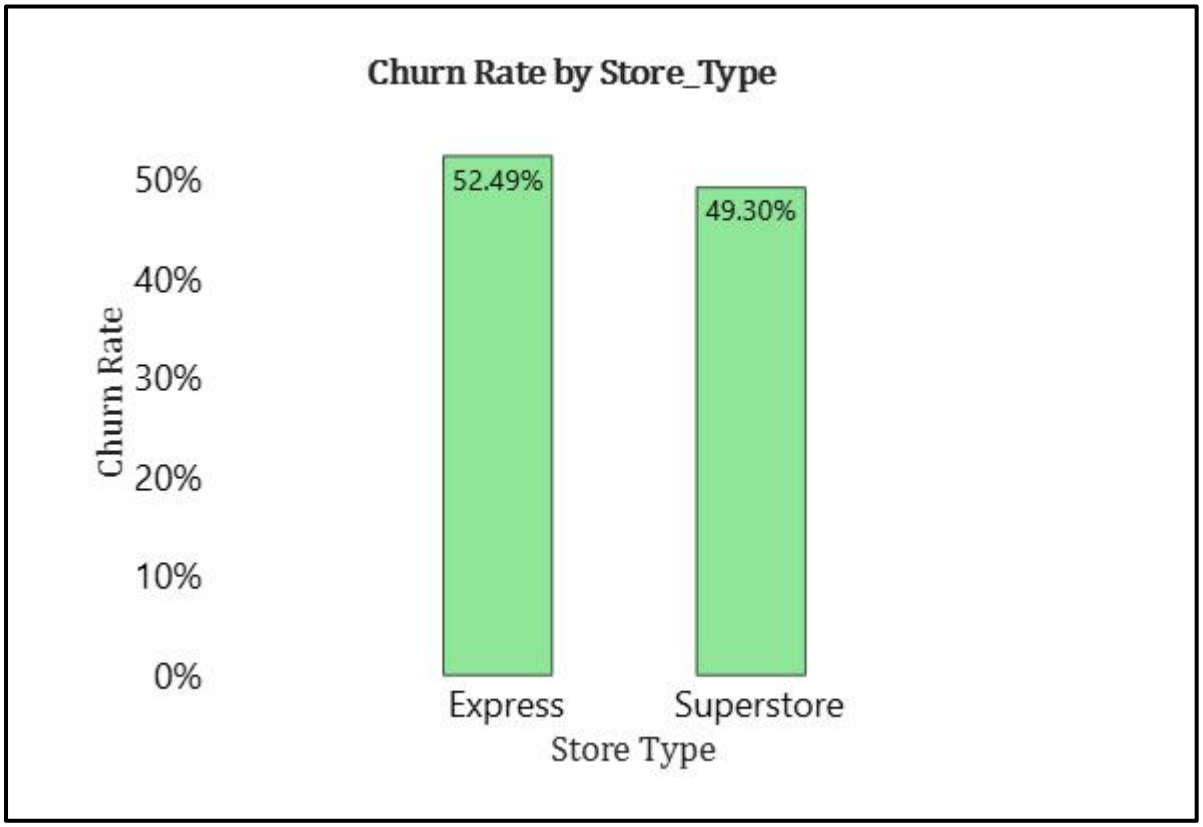


- **<u>To visualize Churn Rate by Income Group –</u>**

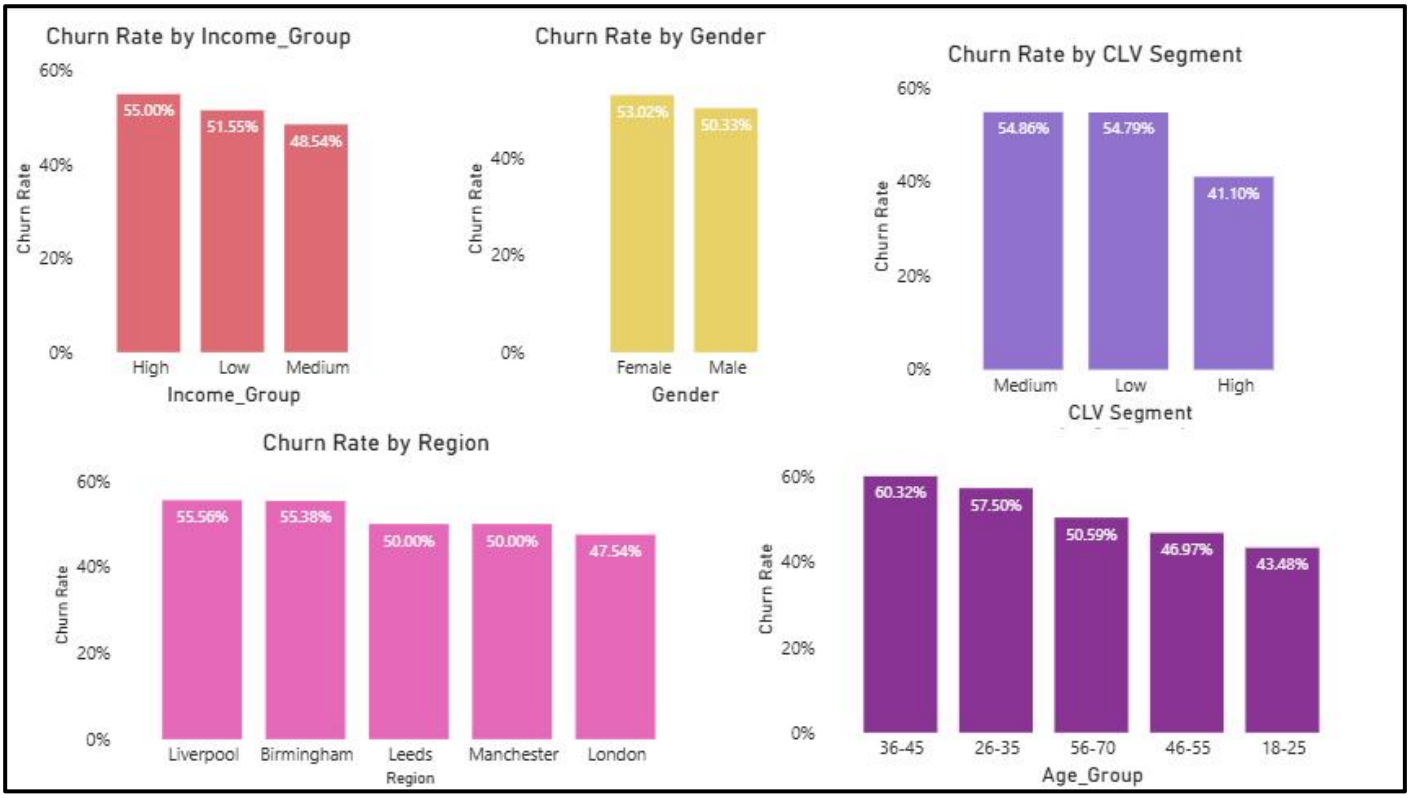


- **<u>To visualize Churn Rate by Store Type –</u>**

As we don't have the common column between **Churn Labelled Customer** table and **Store Location** table so I use **Merge Query** feature in **Power Query Editor** and merges the **Customer Transaction** table and **Store Location** table based on **Store Id** column and get the store type column in **Customer Transaction** Table then I make relationship between **Churn Labelled Customer** table and **Customer Transaction** table and create the chart given below.



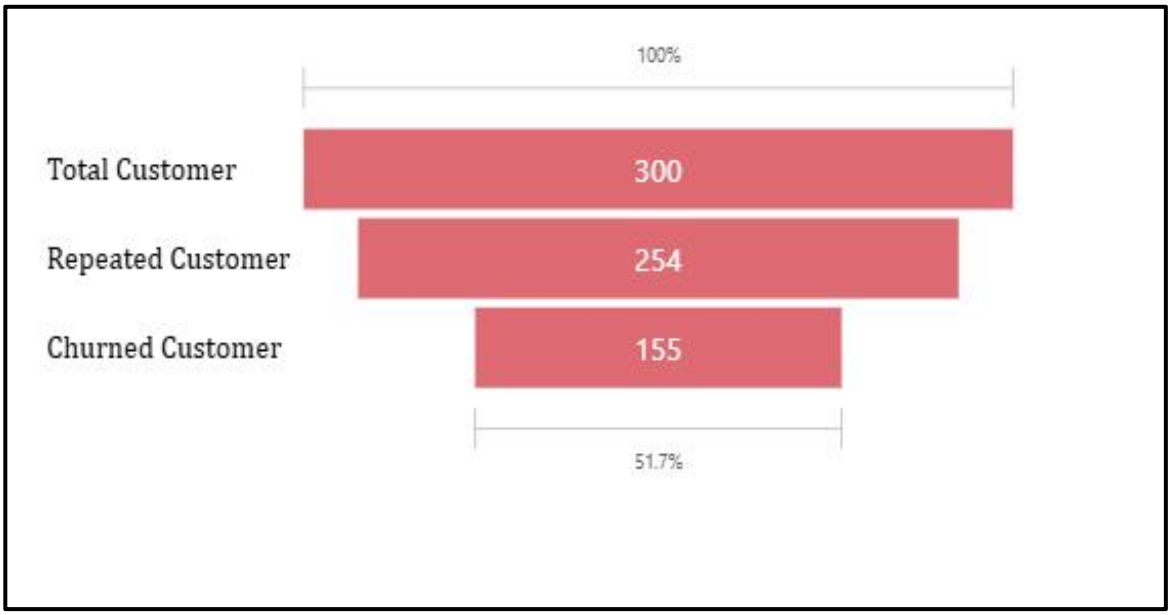
- **<u>Identify Top 5 segments with highest churn % -</u>**



Top 5 segmemts with highest chutn %

1. **Age Group** - (36 - 45)
2. **Region -** Liverpool
3. **Income Group -** High
4. **CLV Segment -** Medium
5. **Gender -** Female

- **<u>Create funnel chart: Total Customers → Repeat → Churned -</u>**



Task 3 : Repeat Purchase Analysis

- **<u>Create a measure:</u>**

Low-Tier Customers: (2-4 purchases)

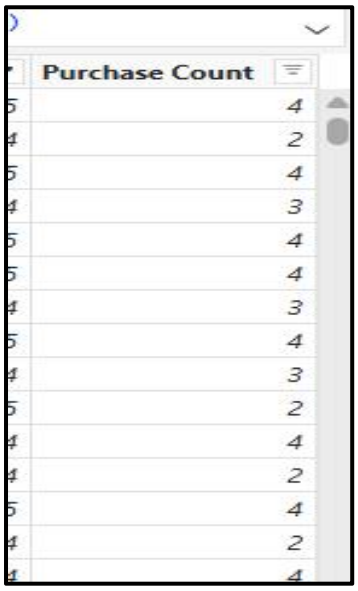
Mid-Tier Customers: (5-10 purchases)

High-Tier Customers: (11+ purchases)

To create these measures first we have to create a column having purchase count so , to create this column right click on **Customer_Transaction** table in Data Pane then choose **New Column** and use DAX –

Purchase_Count=CALCULATE(COUNTROWS(Customer_Transactions),ALLEXCEPT(Customer_Transactions,Customer_Transactions[Customer_ID]))

After that a new column will be added to table.

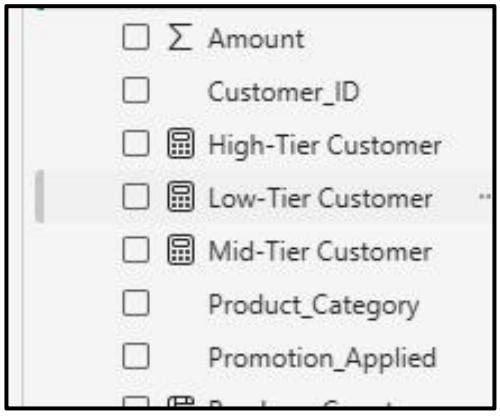


Purchase Count
4
2
4
3
4
4
3
4
3
2
4
2
4
2
4

Now Create measure – To create measure right click on Customer_Transactions table in Data Pane and click on new measure and use the DAX as given below-

1. **Low-Tier Customers** = CALCULATE(DISTINCTCOUNT(Customer_Transactions[Customer_ID]),FILTER(VALUES(Customer_Transactions[Customer_ID]),CALCULATE(COUNTROWS(Customer_Transactions))>=2&&CALCULATE(COUNTROWS(Customer_Transactions))<=4))
2. **Mid-Tier Customers** = CALCULATE(DISTINCTCOUNT(Customer_Transactions[Customer_ID]),FILTER(VALUES(Customer_Transactions[Customer_ID]),CALCULATE(COUNTROWS(Customer_Transactions))>=5&&CALCULATE(COUNTROWS(Customer_Transactions))<=10))
3. **High-Tier Customers** = CALCULATE(DISTINCTCOUNT(Customer_Transactions[Customer_ID]),FILTER(VALUES(Customer_Transactions[Customer_ID]),CALCULATE(COUNTROWS(Customer_Transactions))>=11))

After that all three measure will be added to Data Table in Data Pane.

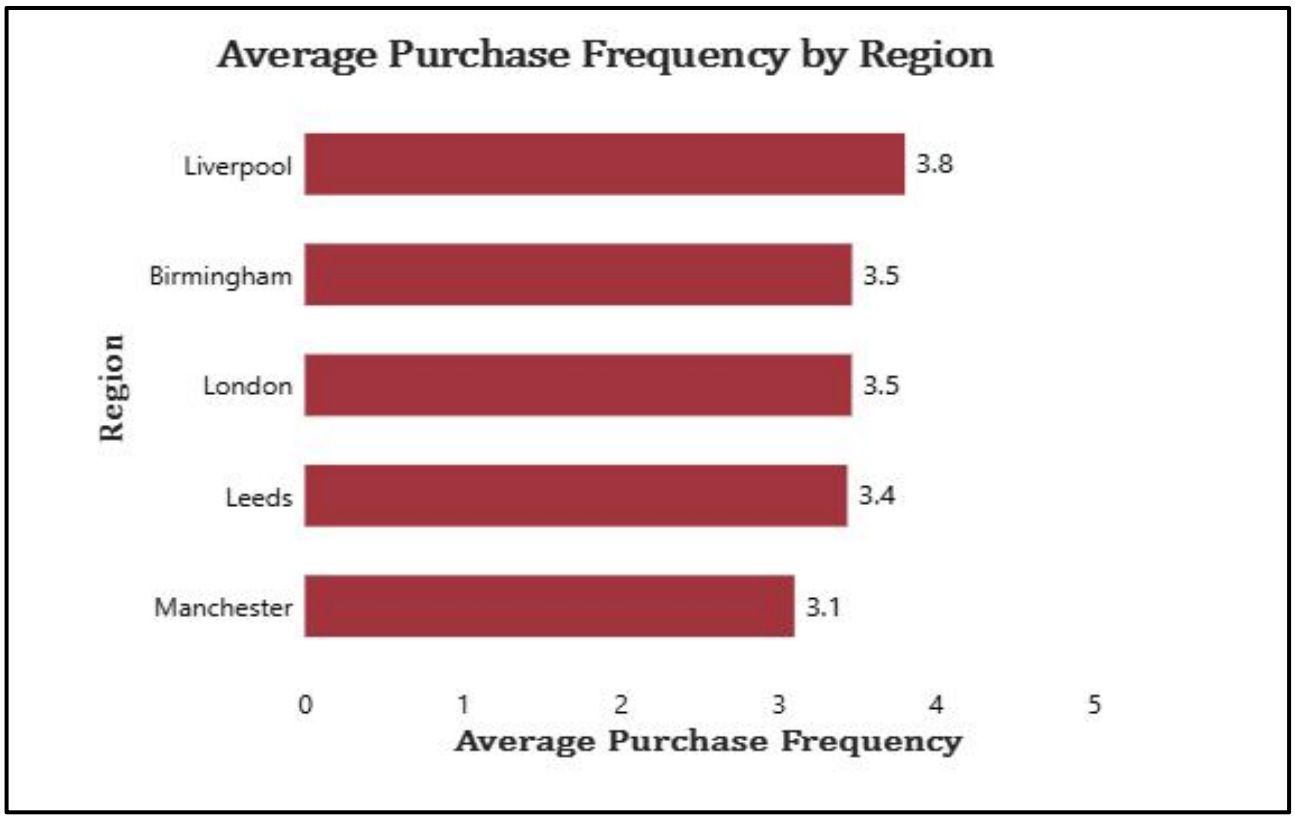


- **<u>Compare avg. purchase frequency by Region –</u>**

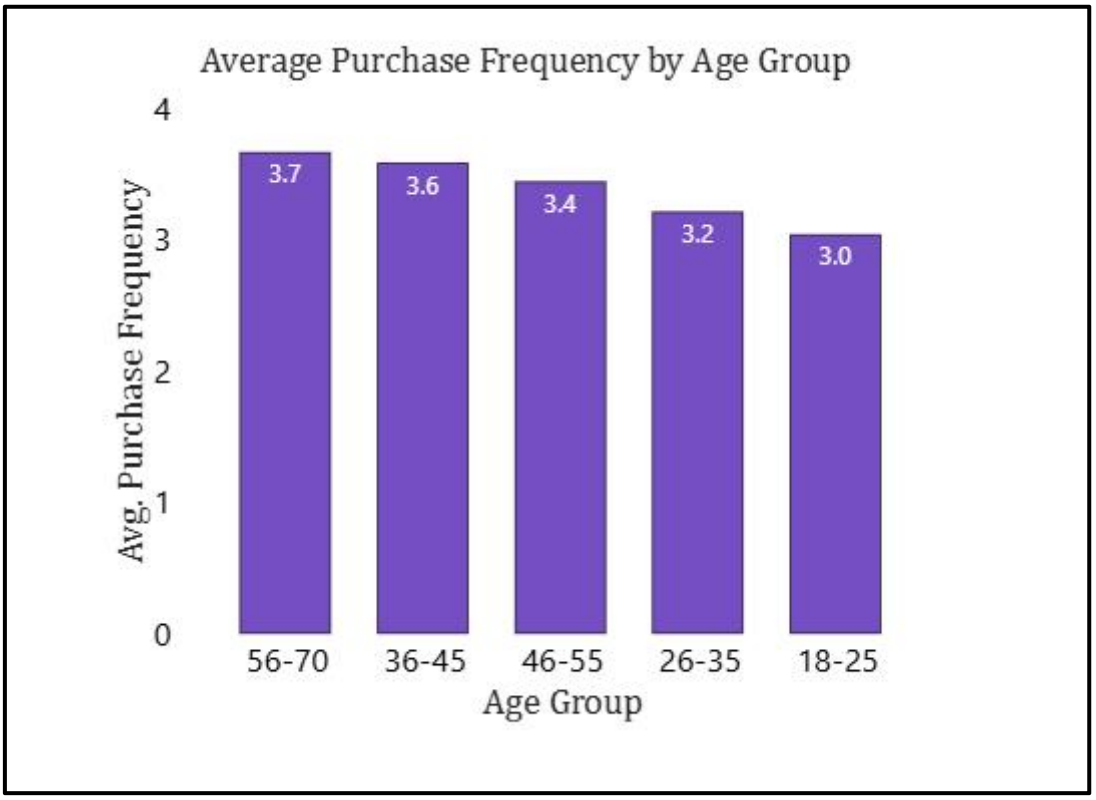
Here I take **Average Purchase Frequency = (Total Purchase) / (Total Customer Who makes Purchase)**

Note – Total Purchase means count of all distinct **Transaction ID** in **Customer Transaction Table.**

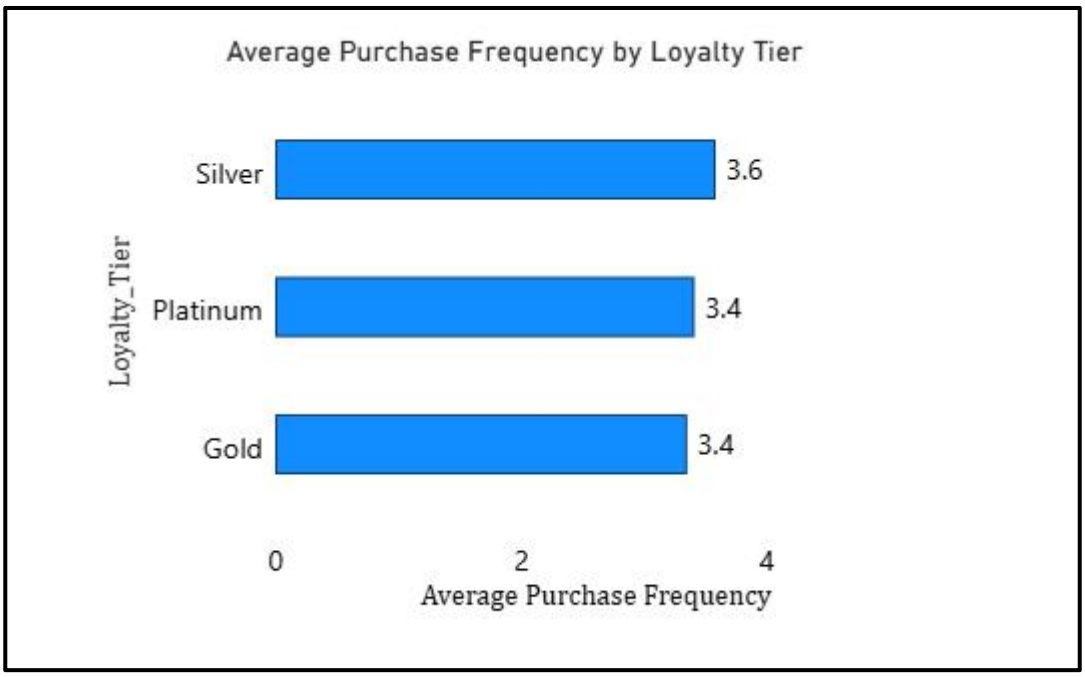
Total Customer means count of all distinct **Customer ID** in **Customer Transaction Table.**



- **Compare average Purchase Frequency by Age Group –**



- **Compare Average Purchase Frequency by Loyalty Tier –**



- **To identify product categories most frequently bought by Loyal Customer -**

Firstly I made a criteria for Loyal Customer that the customers who made four or more purchases are the Loyal Customer.

Conclusion – Most frequently bought product categories by Loyal Customers are **Clothing, Beverages** and **Electronics.**

Task 4 : Promotion And Loyalty Impact

- **Create measure for % transactions where promotion was applied –**

To create the measure right click on Customer_Transactions table in Data Pane and click on New Measure

And write the DAX as follows –

Transaction Where Promotion Applied
=CALCULATE(COUNTROWS(Customer_Transactions),FILTER(Customer_Transactions,Customer_Transactions[Promotion_Applied]="Yes"))

Total Transaction = COUNT(Customer_Transaction[Transaction_ID])

% Transactions Promotion Applied = ([Transactions where promotion applied] / [Total Transaction])*100

After that press Enter and the required measure will be added to Customer_Transaction table in Data Pane.

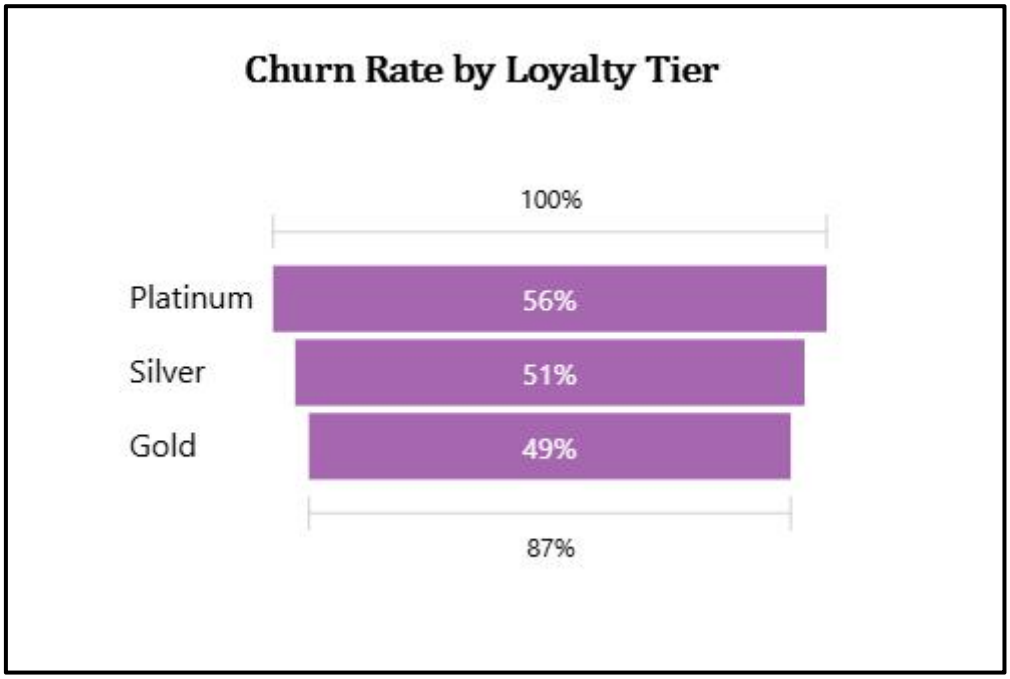
- **To create measure for Avg. purchase amount with vs without promotion –**

To create this measure right click on **Customer_Transactions** table and click on **New Measure** and use the DAX as given below –

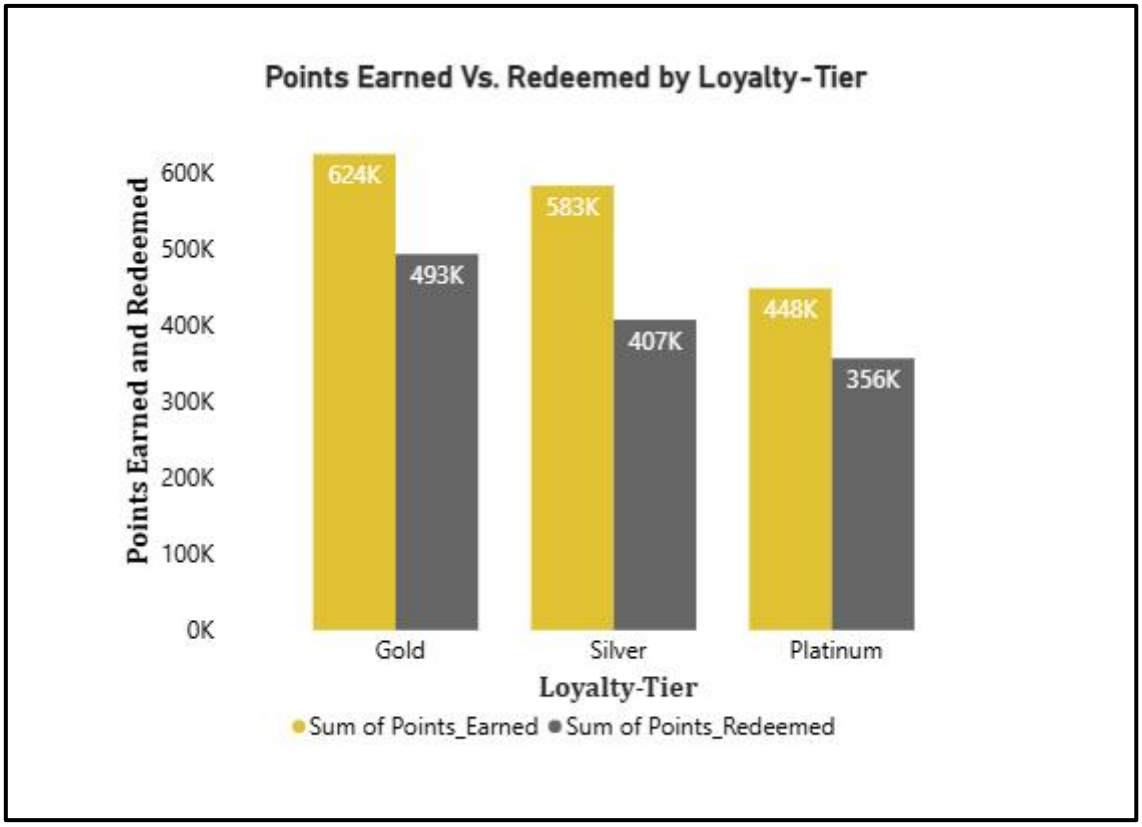
Avg. Amount Where Promotion Applied =
CALCULATE(AVERAGE(Customer_Transactions[Amount]),FILTER(Customer_Transactions,Customer_Transactions[Promotion_Applied]="Yes"))

Avg. Amount Where Promotion is not applied =
CALCULATE(AVERAGE(Customer_Transactions[Amount]),FILTER(Customer_Transactions,Customer_Transactions[Promotion_Applied]="No"))

- **Compare Churn Rate across Loyalty Tier –**



- **<u>Show Points Earned vs Redeemed by Tier using clustered column chart –</u>**



- **<u>Recommend how to improve redemption and retention –</u>**

Recommendation to improve Redemption –

1. **Personalize Offers –** Use customer data and show them selection of rewards based on their preferences and interest.
2. **Push Expiry Reminders –** Send SMS/app notifications before vouchers or offers expire which creates urgency.

3. **Tiered Rewards –** Give better offers or rewards as customer move to higher tiers (eg – silver to gold).
4. **Micro Rewards –** Offer customer small and instant rewards (e.g. free shipping, small cashback) to build redemption habit.

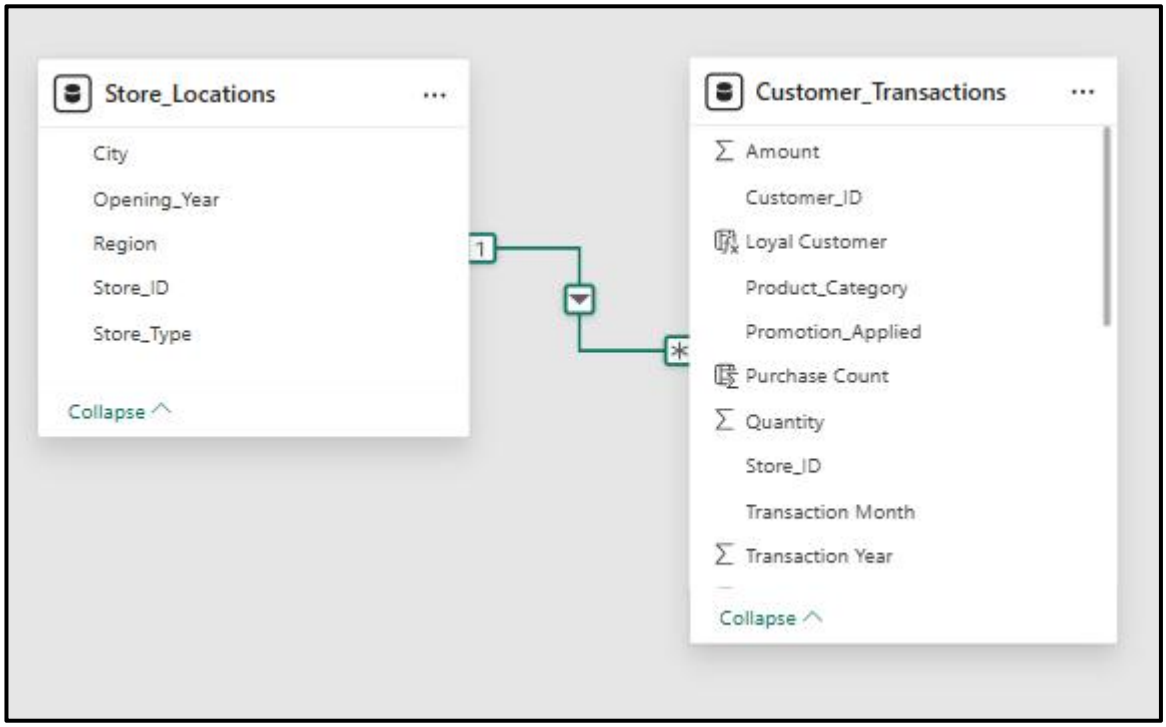
Recommendation to improve Retention –

1. **Engage Repeat Customer –** Reward customers with 2- 4 purchase with some perks to make them loyal.
2. **VIP Treatment** - Give special treatment to customers who makes more purchase such as early access to sales, access to exclusive collections and loyalty points.
3. **Offer Trendy/affordable items** – To reduce churn rate in younger age group offer them trendy and affordable items.
4. **Use Feedback Loop –** Take feedbacks from churn or inactive customers and use them to fix product or service gaps.
5. **Campaign -** Use campaigns in high churn area to make churn or inactive customers aware.

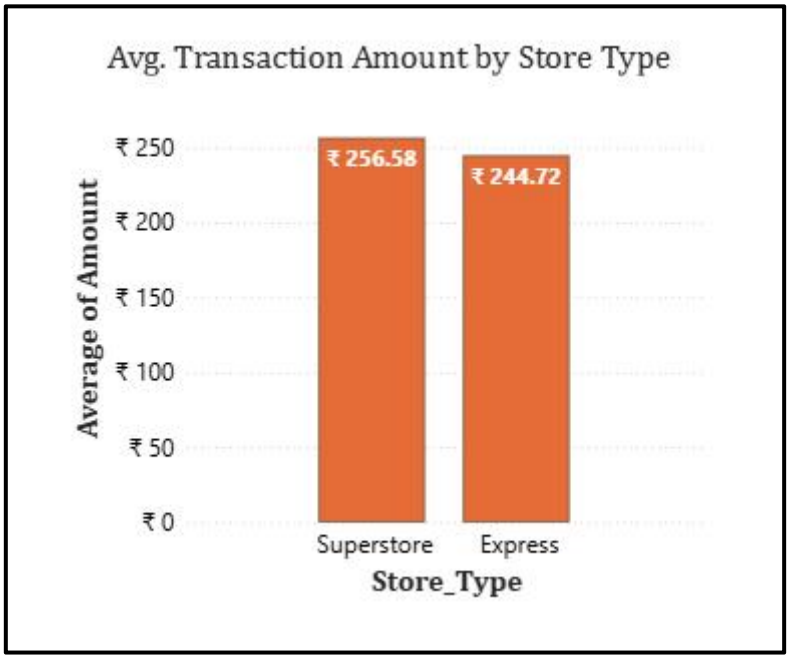
Task 5: Store Performance vs Retention

- **Merge Store_ID from transactions with Store_Locations –**

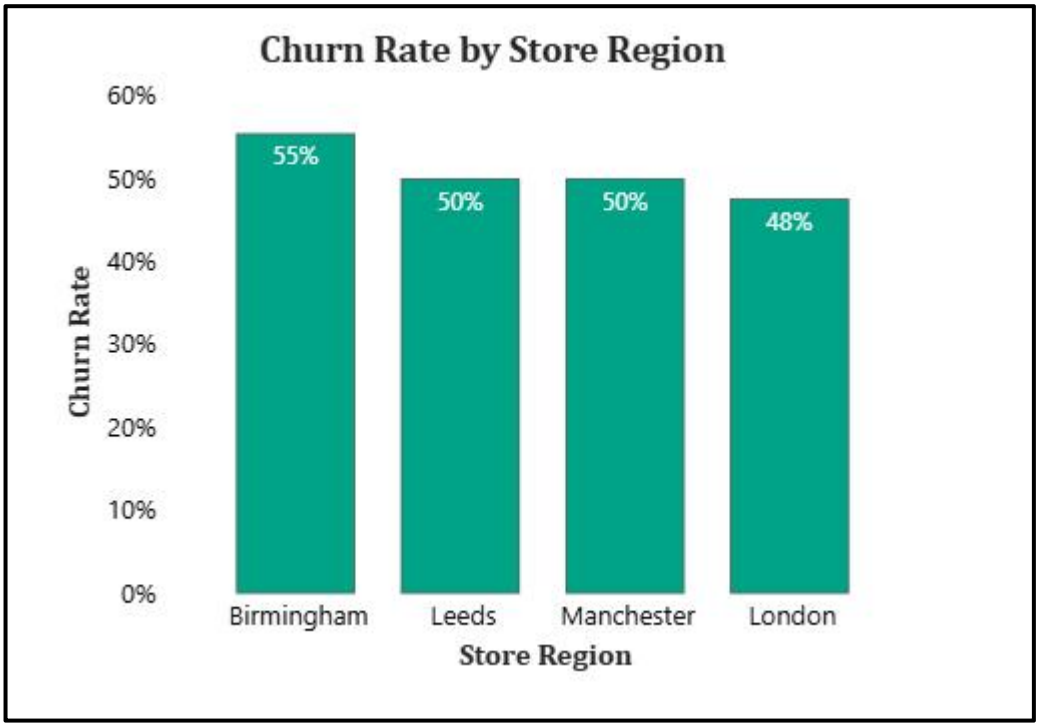
In Power BI we can merge them by making a relationship between them.



- **Visualize: Avg. transaction amount by store type –**



- **Visualize - Churn rate by store region –**



- **<u>Correlation between store opening year and retention –</u>**

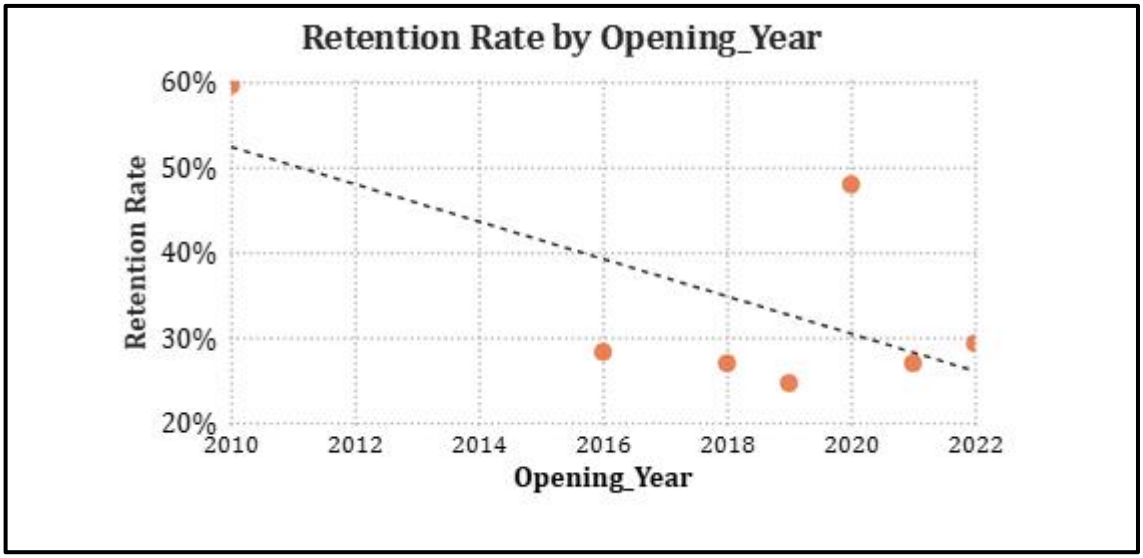
To calculate correlation between store opening year first we create a measure as **Retention Rate.**

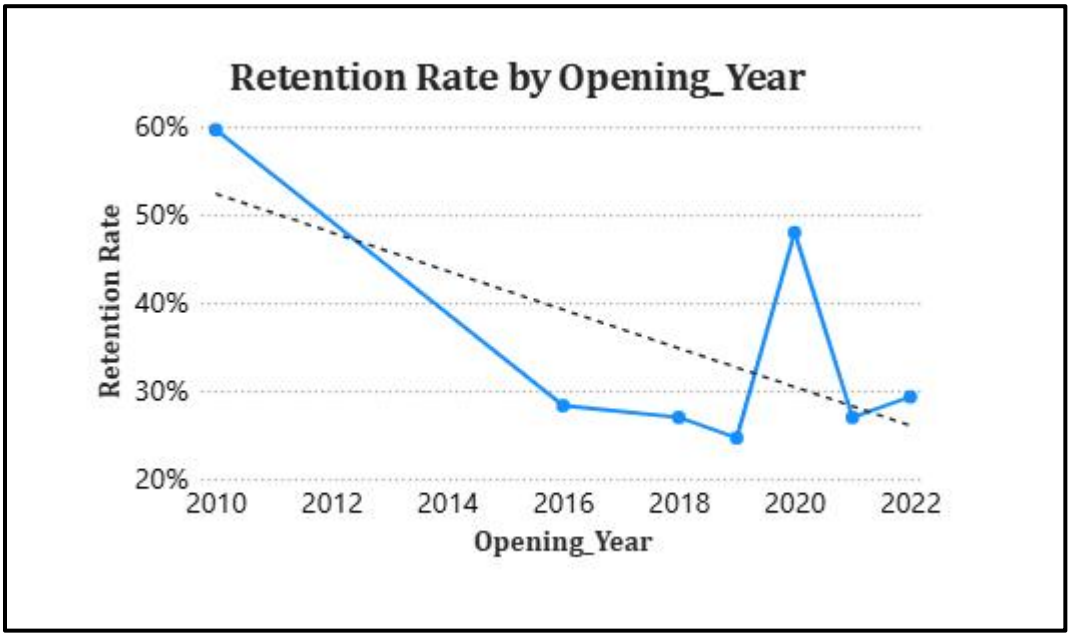
Retained Customers (customer with purchase count 2 or more)
= CALCULATE(DISTINCTCOUNT(Customer_Transactions[Customer_ID]),FILTER(Customer_Transactions,Customer_Transactions[Purchase Count]>=2))

Total Customer = COUNTROWS(Churn_Labelled_Customers)

Retention Rate = [Retained Customers] / [Total Customer]

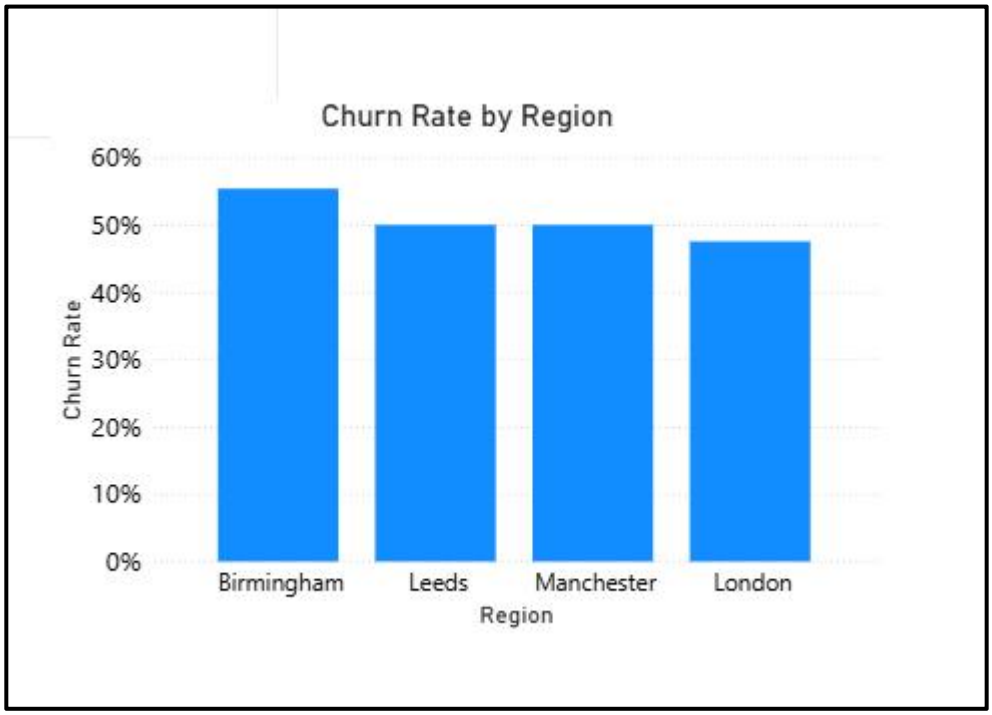
Now we will visualize correlation between store opening year and retention.





As we can clearly see that trendline slopes downward in both chart it means there is **Negative** **Correlation** between Store Opening Year and Retention Rate (retention).

- **Suggest where to run store-specific campaigns –**



Ae in evey store region there is high churn rate so company should run campaign to reduce churn rate in these region.

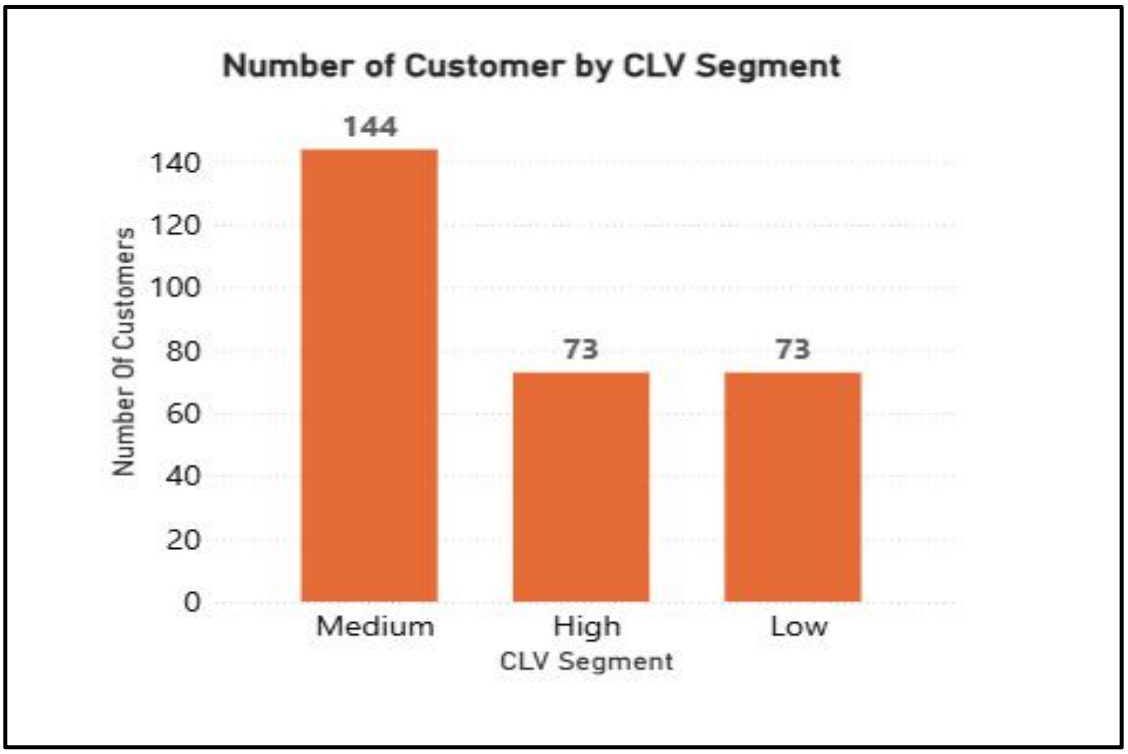
Task 6: Customer Value (CLV) Analysis

● **Segment customers into:**

○ Low (Bottom 25%)

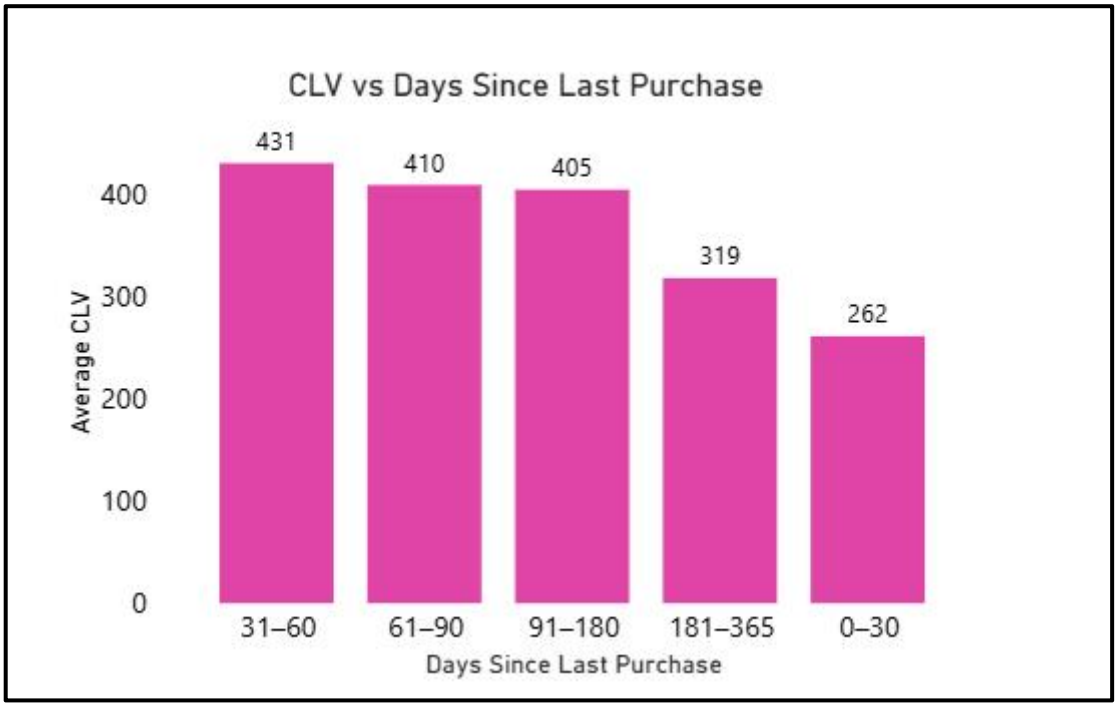
○ Medium (Mid 50%)

○ High (Top 25%)

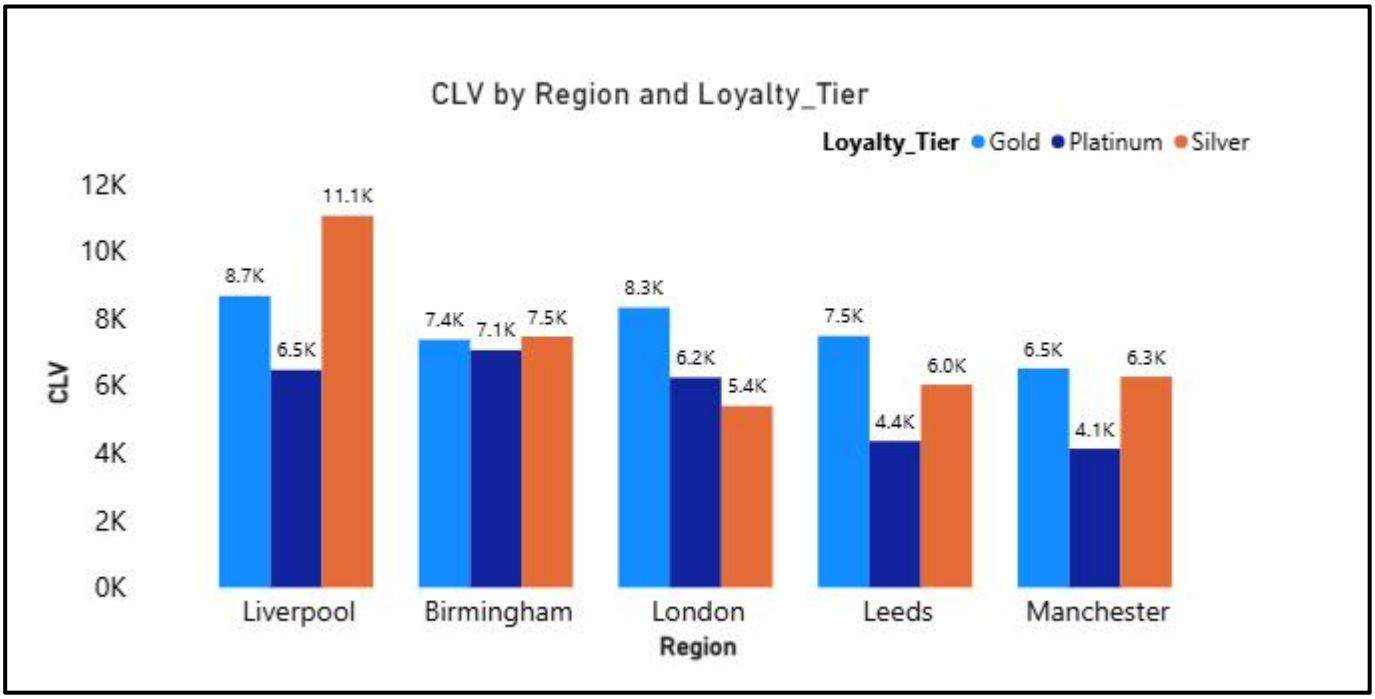


Visualize:

CLV vs Days Since Last Purchase



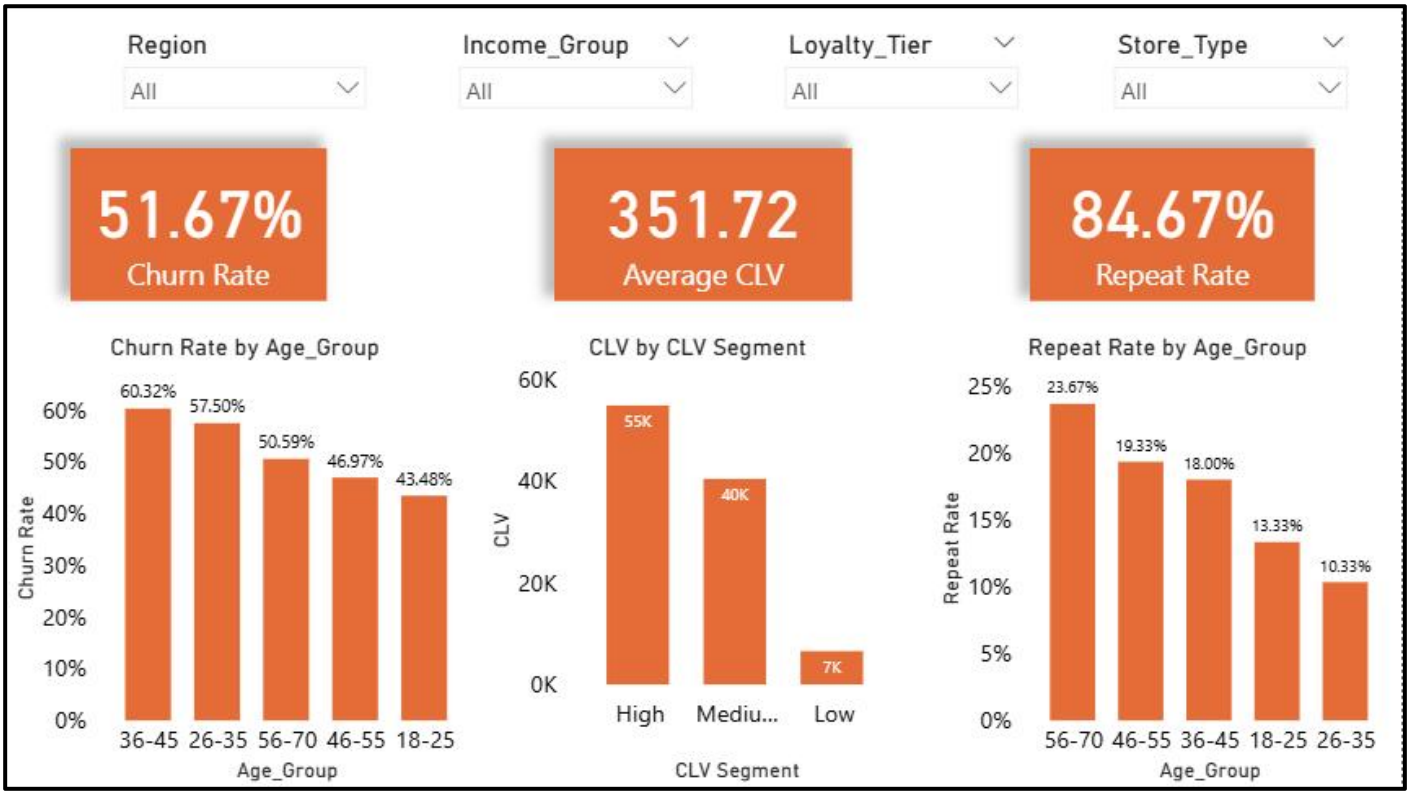
- **<u>CLV by Loyalty Tier and Region</u>**



Task 7: Final Dashboard and Executive summary

Create a multi-page Power BI report:

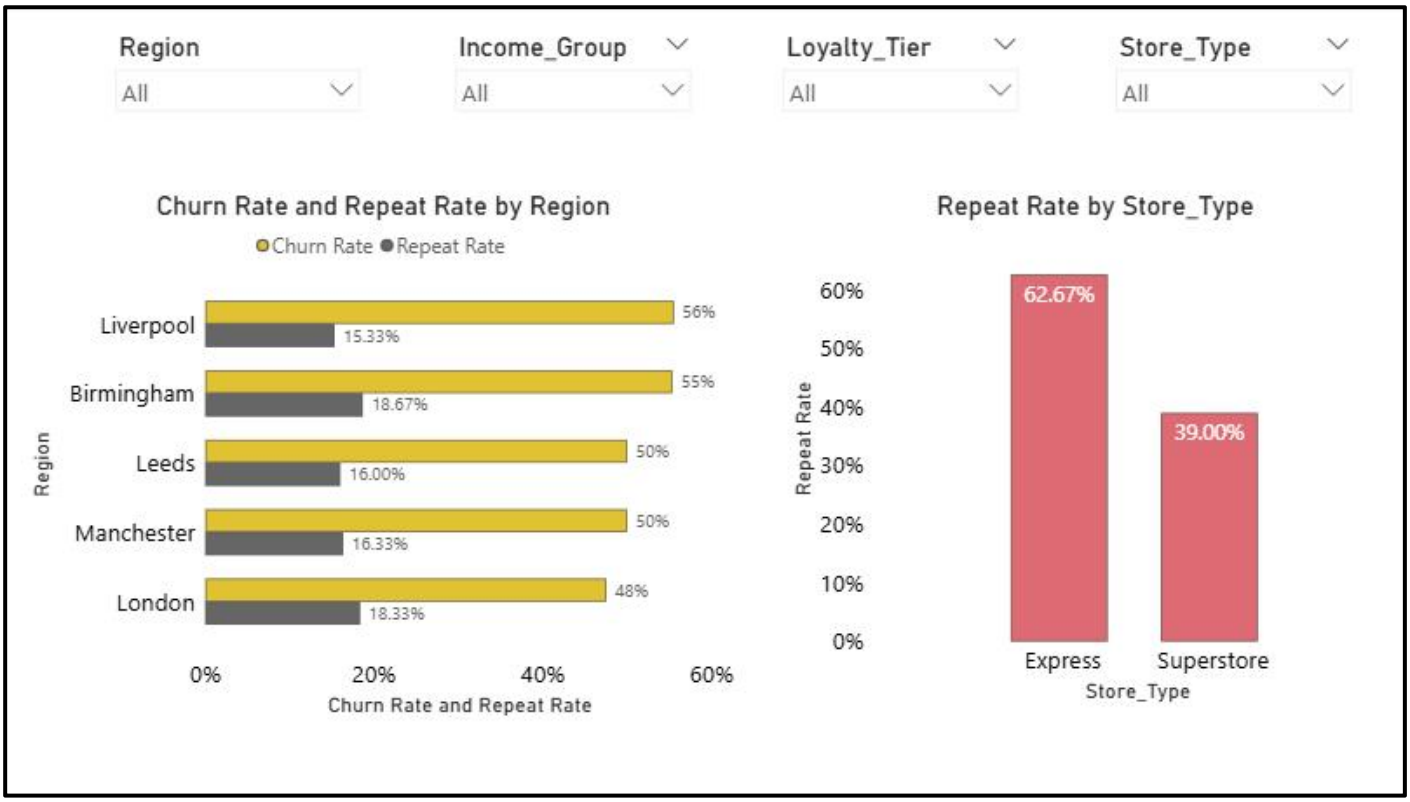
○ **Page 1: Overview KPIs (Churn, CLV, Repeat Rate) -**



○ **Page 2: Loyalty & Promotion Impact -**

As we can there is a mild impact of **Loyality** and there is negligible impact of **Promotion**.

○ **<u>Page 3: Store and Region Insights -</u>**



As we can see that Churn Rate is very high as compare to Retention Rate and for store customers are liked to go to Express Store than Superstore so company should invest more in Express Store and for Super store company should run Campaign.

Page 4: <u>Customer Segmentation (Churned, Repeat, High-Value)</u>

- **Create a card or textbox summarizing top 3 recommendations: -**

- **<u>What should TESCO do to retain more customers?</u>**

1. **Loyalty Programs** - Company should run loyality programs to make customer stay loyal.

2. **Personalised Offers -** On successful purchase company should give some offers or rewards according to customer interest.

3. **Campaign -** Use campaigns in high churn area to make churn or inactive customers aware.

- **Where should they focus next?**

1. Target underperforming stores/regions with localized campaigns.
2. Protect High Value Customer from losing give then VIP offers and rewards.
3. Improve the shopping experience.
4. Run Loyality program and Retention Campaigns.

Project Video Link -

https://www.loom.com/share/a3517f4356a44d6bac4a49801bbacbc9?sid=3bbaccf3-c98f-445f-82a1-c563799bbed4